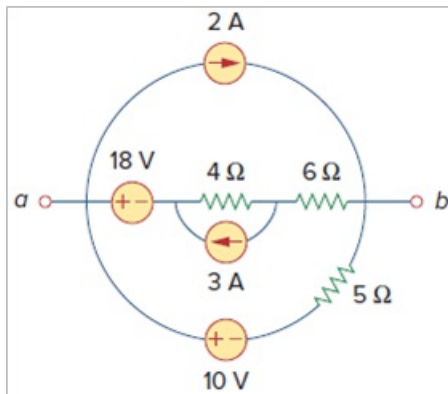


Problem

For the circuit in Fig. 4.126, find the Thevenin and Norton equivalent circuits at terminals a - b .

Figure 4.126:



Step-by-step solution

Step 1 of 7

Refer to the circuit diagram depicted in Figure 4.126 in the textbook.

In Figure 4.126, in the middle limb, there is parallel combination of current source with resistor. Apply source transformation and convert it into equivalent voltage source in series with resistance. Draw the modified circuit diagram.

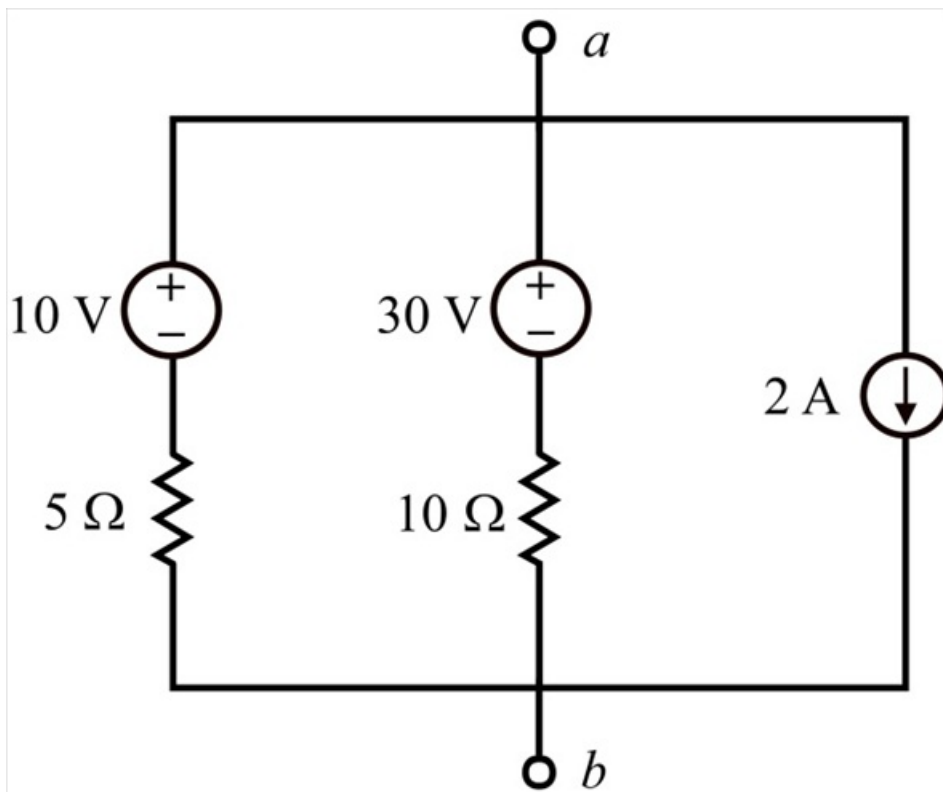


Figure 1

Step 2 of 7

To find the Thevenin equivalent resistance, R_{Th} , short circuit the voltage sources and open circuit the current source in Figure 1. Re-draw the modified circuit diagram.

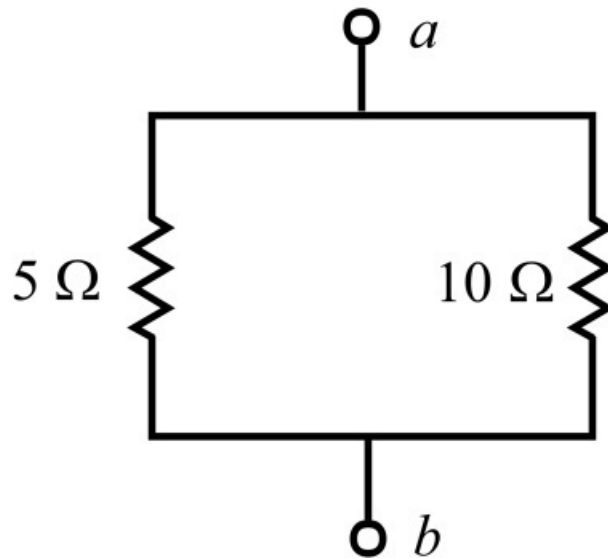


Figure 2

Step 3 of 7

Determine the Thevenin resistance from Figure 2.

$$\begin{aligned} R_{Th} &= 5 \parallel 10 \\ &= \frac{5 \times 10}{5 + 10} \\ &= 3.333\ \Omega \end{aligned}$$

Therefore, the Thevenin resistance is, $3.333\ \Omega$.

Step 4 of 7

In Figure 1, assume that the potential at terminal a is V_a and terminal b is at ground potential.

Apply Kirchhoff's current law at node V_a .

$$\begin{aligned} \frac{V_a - 10}{5} + \frac{V_a - 30}{10} + 2 &= 0 \\ V_a \left(\frac{1}{5} + \frac{1}{10} \right) &= 2 + 3 - 2 \\ V_a (0.3) &= 3 \\ V_a &= 10\ \text{V} \end{aligned}$$

Determine the Thevenin voltage at terminals a and b .

$$\begin{aligned} V_{Th} &= V_{ab} \\ &= V_a - V_b \\ &= 10 - 0 \\ &= 10\ \text{V} \end{aligned}$$

Therefore, the Thevenin voltage is $10\ \text{V}$.

Step 5 of 7

Draw the Thevenin equivalent circuit.

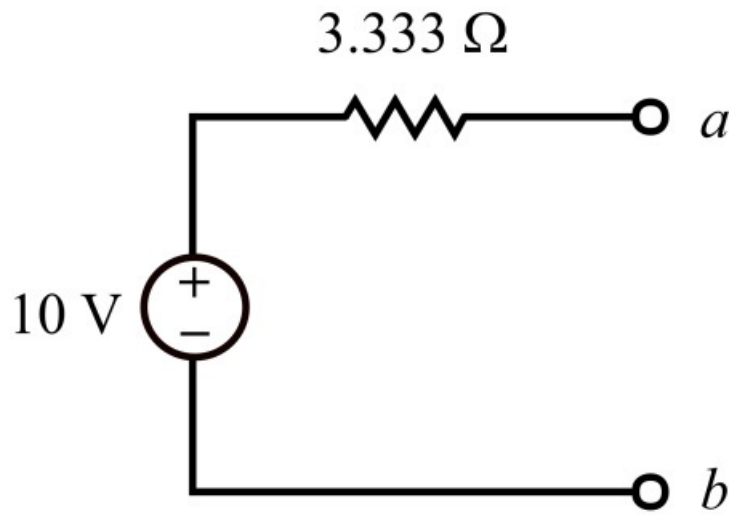


Figure 3

Step 6 of 7

The Norton resistance is same as the Thevenin resistance. Therefore,

$$\begin{aligned} R_N &= R_{Th} \\ &= 3.333 \, \Omega \end{aligned}$$

Determine the Norton short circuit current.

$$\begin{aligned} I_N &= \frac{V_{Th}}{R_N} \\ &= \frac{10}{3.333} \\ &= 3 \, \text{A} \end{aligned}$$

Step 7 of 7

Draw the Norton equivalent circuit.

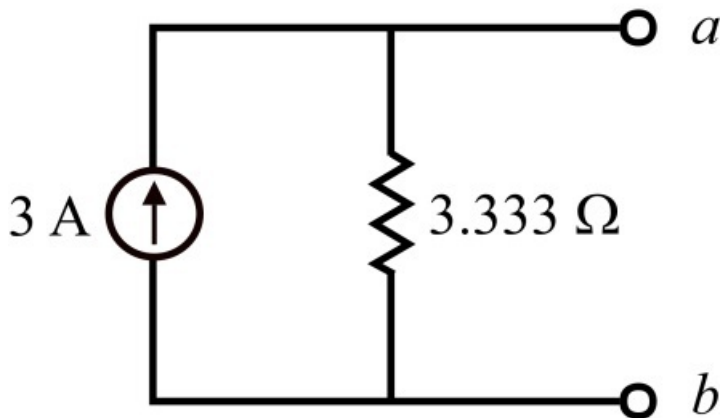


Figure 4